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| **Fiscal Year:** | FY 2026 | **Task Last Updated:** | FY 01/24/2026 |
| **PI Name:** | Yagoobi, Jamal Ph.D. | | |
| **Project Title:** | Electrically Driven Liquid Film Flow Boiling: A Two-Phase Heat Transport Device Driven By Electric Conduction Mechanism | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | UNIVERSITY | **Phone:** | 508-831-5759 |
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| **City:** | Worcester | **State:** | MA |
| **Zip Code:** | 01609-2280 | **Congressional District:** | 3 |
| **Comments:** | NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012) | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | |
| **Start Date:** | 03/01/2022 | **End Date:** | 02/28/2027 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA GRC |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | NOTE: End date changed to 02/28/2027 per F. Hernandez/NASA (Ed., 4/8/26). | | |
| **Key Personnel Changes/Previous PI:** | Not applicable | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | 80NSSC22K0676 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | NOTE: Task Description updated from original proposal (Ed., 7/18/23)<br>The basic science and experimental effort covered by NASA Grant/Cooperative Agreement Number: NNX16AT09G entitled, “Electrically Driven Liquid Film Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism” is essentially continuing with this new proposal. The specific tasks of the proposed project are:<br><br>Task 1: Support the final design and analysis of the experiment conduction disc and heater, as well as detailed thermal analysis of the thermodynamics governing the experiment chamber. Develop and conduct ground-based experiment to determine the final flight matrix for potential study on the sub-orbital flights or International Space Station (ISS). | | |

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| **Task Description:** | Task 2: Complete tests as stated in the Science Requirements Document (SRD) (period: October 2021 to December 2022), with 2 mm liquid film thickness. The same will be conducted with the liquid film thickness of 3 mm. Upon completion of this effort, the combined results of 2 mm and 3 mm tests will used to finalize the test matrix to be conducted potentially aboard the sub-orbital flights or the ISS.<br><br>Task 3: Continue with investigating the impact of the presence of non-condensable gas. The dielectrophoretic (DEP) force extracts the vapor bubble away from the heat surface. This is because of the difference between the vapor phase and liquid phase electric permittivity values. The vapor with a lower electric permittivity is attracted toward a lower electric field (i.e., DEP electrode) and the liquid with a higher electric permittivity is pushed toward the higher electric field (i.e., the heater). In the presence of a non-condensable gas (i.e., air), the DEP will extract the air away from the heater surface, thus minimizing or removing the impact of non-condensable gas on the liquid/vapor phase change taking place at the heater surface. A series of tests were conducted at Worcester Polytechnic Institute (WPI) by introducing air at different levels in 2023 and 2024. More detailed fundamental (theoretical and numerical) and applied work (experimental) in this area will be carried out in 2025 and 2026. In the experiments, the corresponding Psat and Tsat are measured to determine the amount of air introduced into the test vessel. In conclusion, the DEP force not only enhances the boiling and makes it feasible in the absence of gravity; it is also expected to eliminate the negative impact of the presence of non-condensable gas in the working fluid.<br><br>Task 4: Numerical modelling of electrically driven liquid film flow in the presence of phase change in the radial configuration. The electrically driven liquid film flow generated by the electrohydrodynamic (EHD) conduction mechanism will be mathematically modeled to provide scientific results for the corresponding flow field and electric field. The problem in hand is very unique and challenging since the liquid film thickness changes in the direction of the flow due to condensation taking place at its liquid/vapor interface. Furthermore, the downstream boundary condition will need to take into account the presence of boiling and the DEP vapor extraction force. The numerical results will be compared in the presence and absence of gravity.<br><br>Task 5: Detailed flow field measurements of electrically driven liquid film flow. The electrically driven liquid film flow field will be measured with the particle imaging velocimetry (PIV) unit. This will provide valuable data which will be used to compare them to the numerical predictions stated in Task 3. Initial PIV experiments will be conducted using our linear two-phase flow setup. Afterwards, modifications will be made to the radial unit, followed by corresponding flow field measurements with the PIV unit.<br><br>Task 6: A detailed long-term study will be conducted to evaluate the system performance with respect to time and make sure that the effects of EHD conduction pumping of liquid film and the extraction of vapor bubbles do not degrade over time. Tentatively, a period of a full year will be considered for this continuous operation study.<br><br>Task 7. The impact of oscillating the EHD conduction pumping on the heat transport capacity of the electrically driven liquid film flow boiling will be investigated. It is expected that there will be a critical frequency which will result in optimal operating conditions. This oscillation frequency will be related to the electric charge relaxation time.<br><br>Task 8. Determine the impact of presence of electrical field on the performance of electronics during their cooling that is enhanced with externally imposed electrical field. This is essential in order to understand if the performance of the CPUs/chips will be impacted by the externally imposed electric field which is necessary for the operation of the EHD pumping and the activation of dielectrophoretic mechanism for extraction of vapor bubbles during boiling. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The rapid development of modern electronics has resulted in significant computational power in increasingly small packaging. However, a significant challenge remains in effectively controlling the temperature of electronic components, especially in harsh environments including space. With the revitalized interest in space development through NASA's Artemis Program and the increasing commercialization of space, development of improved thermal management technologies is needed with the objectives of reducing the mass and increasing the efficiency of heat transfer systems. Electrohydrodynamic (EHD) pumps have been identified as an advanced technology to support NASA's future human, robotic, and deep space missions. Electrohydrodynamic conduction pumps in particular have the advantage of being lightweight, with no moving parts, and are highly effective for thermal enhancement.<br>From the progress made in this project, the body of knowledge and understanding of EHD conduction pumping applied for thermal management of electronics on earth as well as in space has been expanded. Though this research is targeted for electronics cooling in space, utilization of EHD conduction pumping for enhancement of existing terrestrial systems, such as vapor-compression cycle refrigeration, has the potential for improving the overall system efficiency. There is also the potential to improve the thermal performance of electronics, such as those in data centers found throughout the world. Thus, the future implementation of EHD has the potential for significant cost savings, and reduction in carbon footprint.<br><br>The research conducted in this project has clearly revealed that the Dielectrophoresis (DEP) vapor extraction mechanism plays a critical role in making boiling feasible to cool electronics and electrical devices in zero- and multi-gravity environments. The experiment conducted aboard the Blue Origin Shepard spacecraft in September 2025 clearly confirmed this finding. |
| **Task Progress:** | The research conducted in this project has clearly revealed that the Dielectrophoresis (DEP) vapor extraction mechanism plays a critical role in making boiling feasible to cool electronics and electrical devices in zero- and multi-gravity environments. The experiment conducted aboard the Blue Origin Shepard spacecraft in September 2025 clearly confirmed this finding.<br>a- Dielectrophoresis (DEP) offers a mechanism for phase-change separation of liquid and vapor for enhancement of heat transfer. However, the effect of this mechanism had not been examined on electronic devices' performance previously. This study addressed the performance of electronic cooling with the aid of DEP mechanism. This was illustrated with the use of Raspberry Pi 5 by cooling the CPU surface in three orientations. Results showed that the use of DEP for vapor extraction successfully decreased the CPU surface temperature – as well as maintained, and in some cases improved – the performance and functionality of the Raspberry Pi 5. These findings suggest that DEP can be integrated into thermal management systems to ensure that electronic devices remain at operable temperatures with reliable |

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| | computation performance.<br><br>b- An experiment was carried out aboard the Blue Origin New Shepard spacecraft in September 2025. The flight took place at the Blue Origin flight site located in Van Horn, Texas. The results of this work are currently being analyzed. However, the preliminary analysis confirms that the DEP vapor extraction force successfully removed the vapor bubbles away from the heated surface under zero- and multi-gravity environments. Thus, the presence of DEP mechanism makes boiling possible in space and beyond. |
| **Bibliography Type:** | Description: (Last Updated: 01/30/2026) |
| **Abstracts for Journals and Proceedings** | Podlaski L, Yagoobi J. "Impact of dielectrophoresis force on electronics performance." Joint Electrostatics Society of America Annual Meeting, St. Catharines, Canada, June 22-26, 2025.<br>Abstracts. Joint Electrostatics Society of America Annual Meeting, St. Catharines, Canada, June 22-26, 2025. , Jun-2025 |
| **Awards** | Podlaski L. "Joint Electrostatics Society of America Annual Meeting, second place award for presentation, June 2025." Jun-2025 |